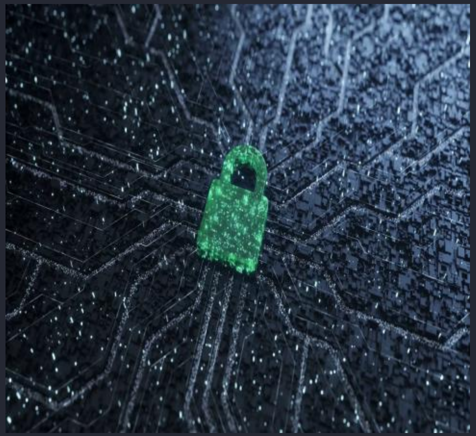


End-to-End Business Intelligence Solution

Fraud Detection Analysis

Uncovering illicit activities through advanced data analytics.

Key Technologies and Tools

Data Generation

Synthetic transaction data for Analysis.

Database & Data Warehouse

SQL Server for transactional data, Star Schema and Role-Playing Dimensions for scalable analytics.

ETL: SSIS

Microsoft SQL Server Integration Services for efficient data extraction, transformation, and loading.

Visualization: Power BI

Interactive dashboards and reports for actionable insights and real-time monitoring.

The Fraud Detection Imperative

Fraud poses a significant and escalating risk to financial institutions globally. The ability to accurately detect and monitor fraudulent transactions is paramount for maintaining financial integrity and customer trust.

- *Financial institutions face substantial risks from evolving fraud*
- *The need for precise fraud transaction detection and monitoring is critical.*
- *Challenges include real-time detection, continuous model evaluation, and comprehensive reporting.*
- *Our goal is to deliver a robust BI solution capable of analyzing fraud patterns, assessing system performance, and tracking key metrics.*

Project Objectives: Building a Resilient Solution

Develop Scalable Data Infrastructure

Implement robust databases and data warehouses to handle large volumes of transaction data efficiently.

Automate ETL Processes

Design and deploy SSIS packages for automated, reliable, and timely data extraction, transformation, and loading.

Enable Real-time Monitoring

Create Power BI dashboards for real-time visualization of fraud alerts, key performance indicators, and trend analysis.

EDA

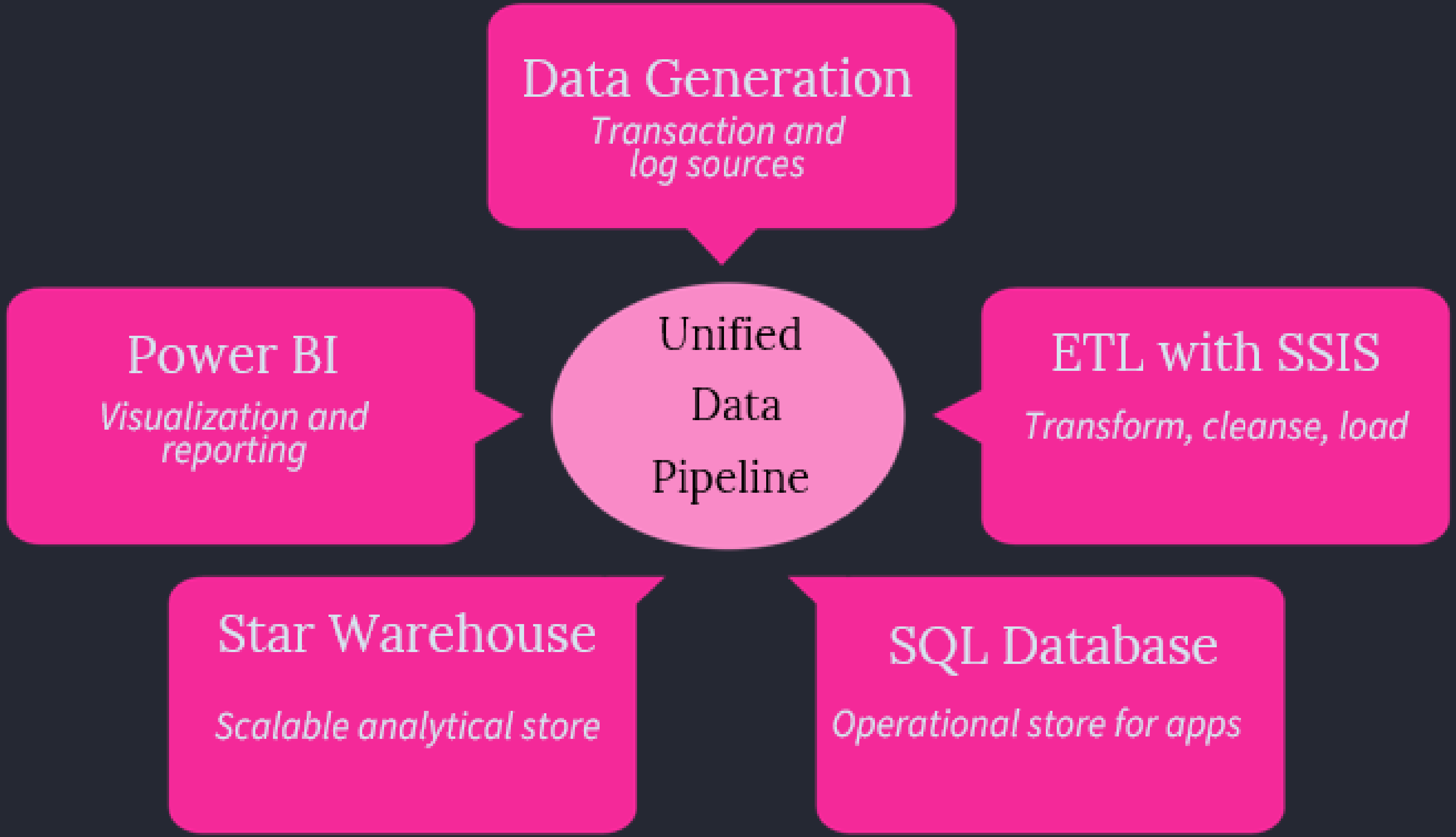
Exploratory Data Analysis using SQL

Provide Actionable Insights

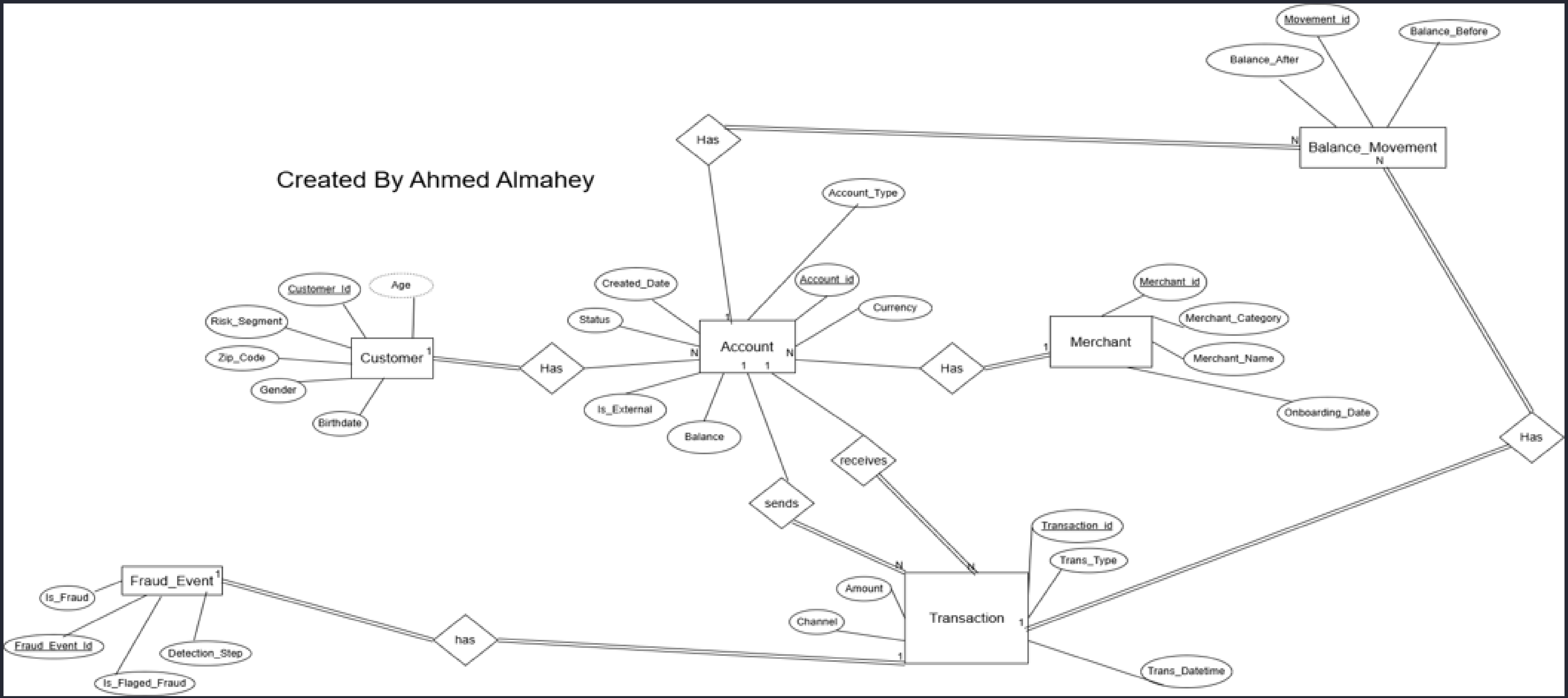
Deliver comprehensive reports for data engineers, analysts, and stakeholders to make informed decisions.

System Architecture: A Unified Approach

Our system architecture provides a holistic view of the fraud detection process, from data ingestion to advanced analytics and visualization.



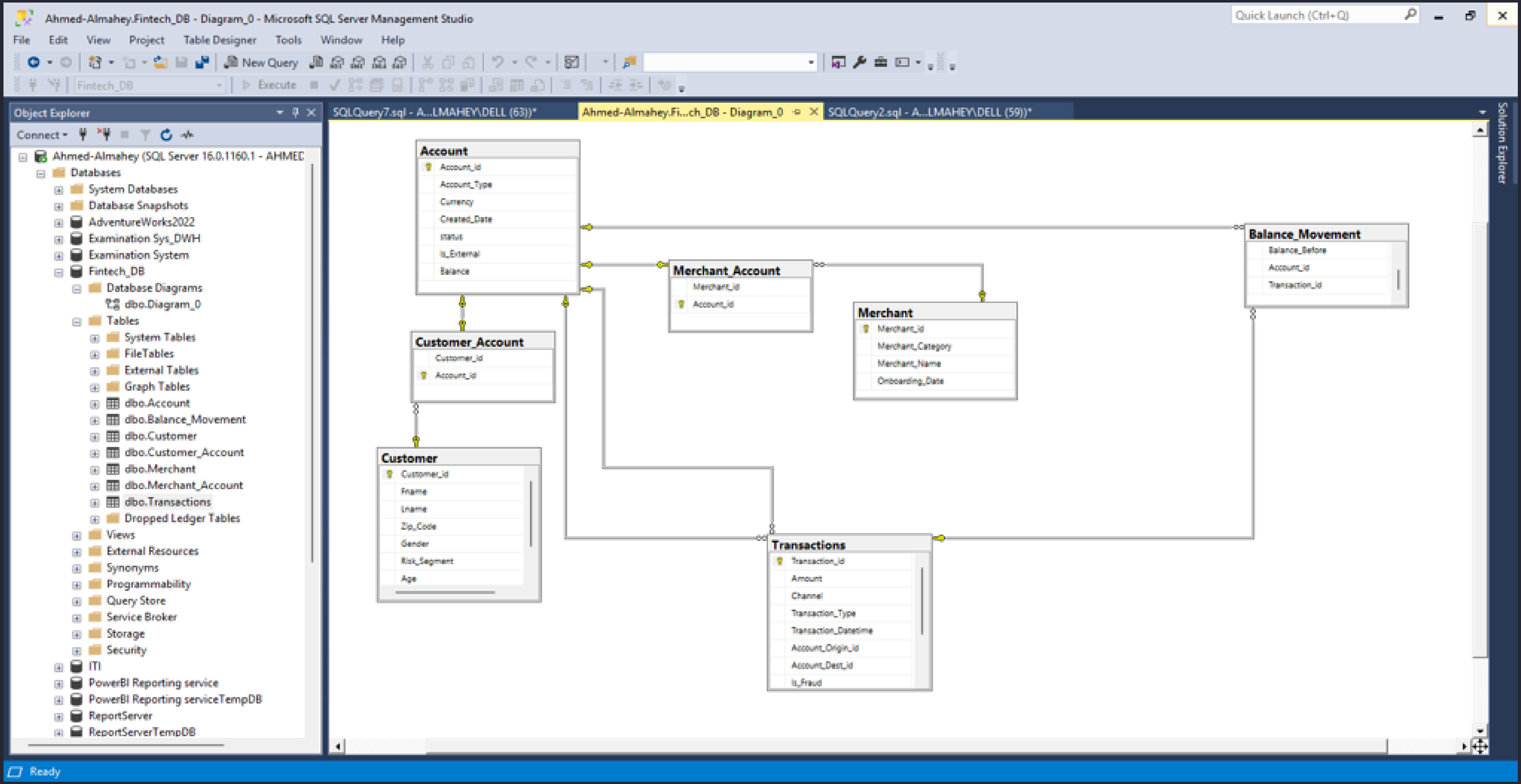
This design ensures data integrity, scalability, and seamless integration across all components, optimizing the fraud detection pipeline.



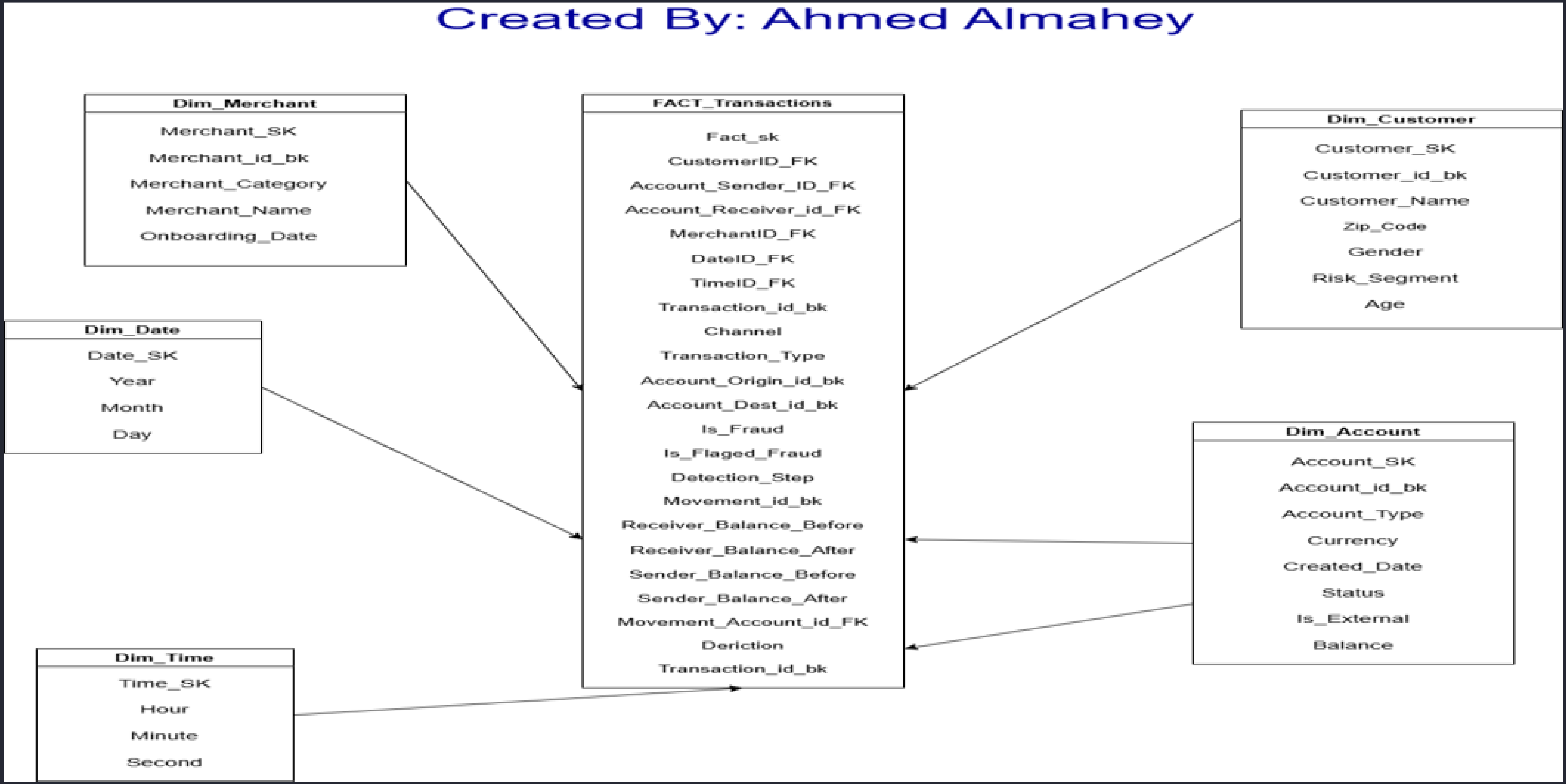
Entity-Relationship Diagram: Foundation of Our Data Model

The ER diagram illustrates the relationships between critical entities in our fraud detection system, ensuring data consistency and integrity.

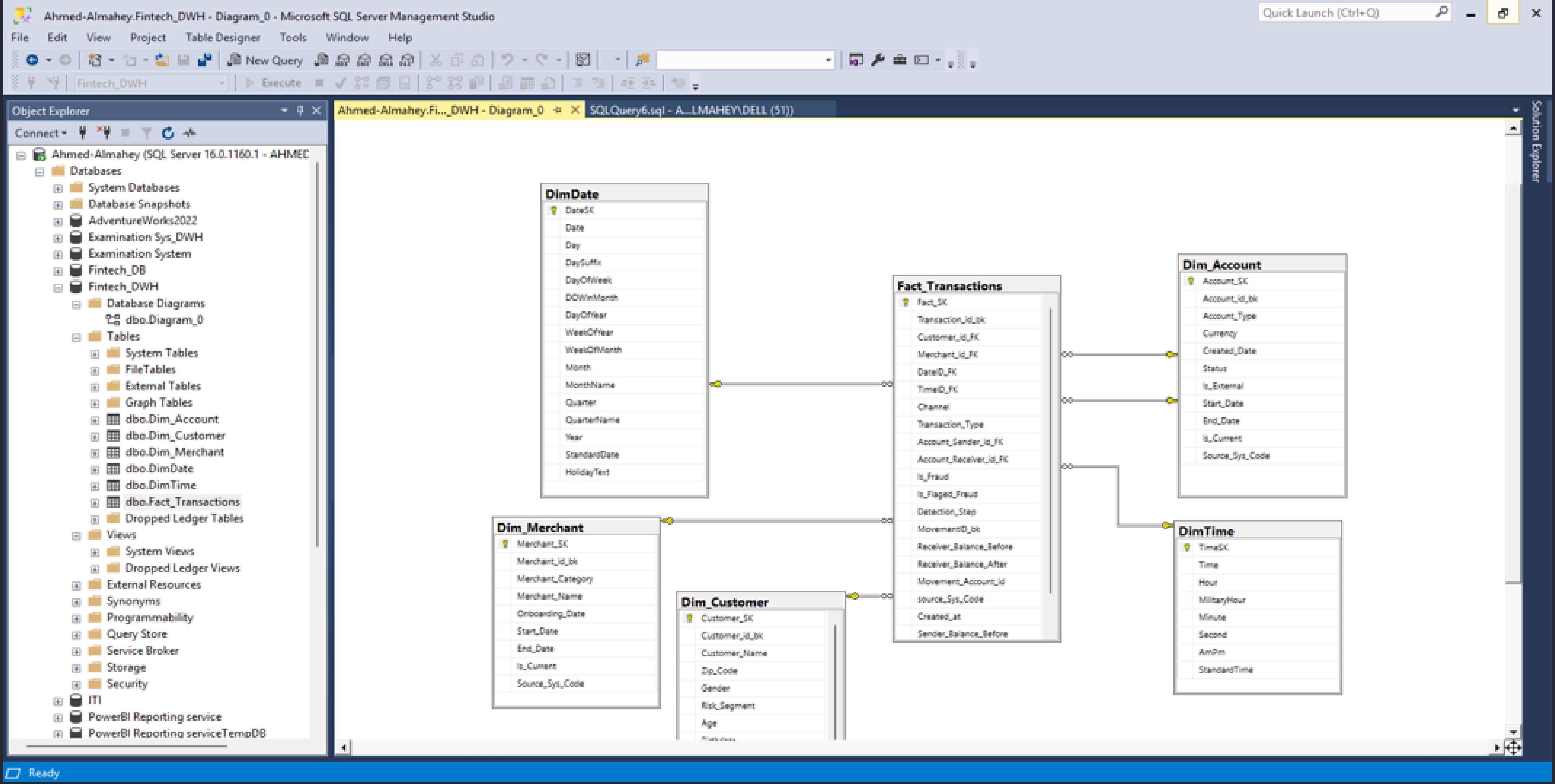
Database Schema



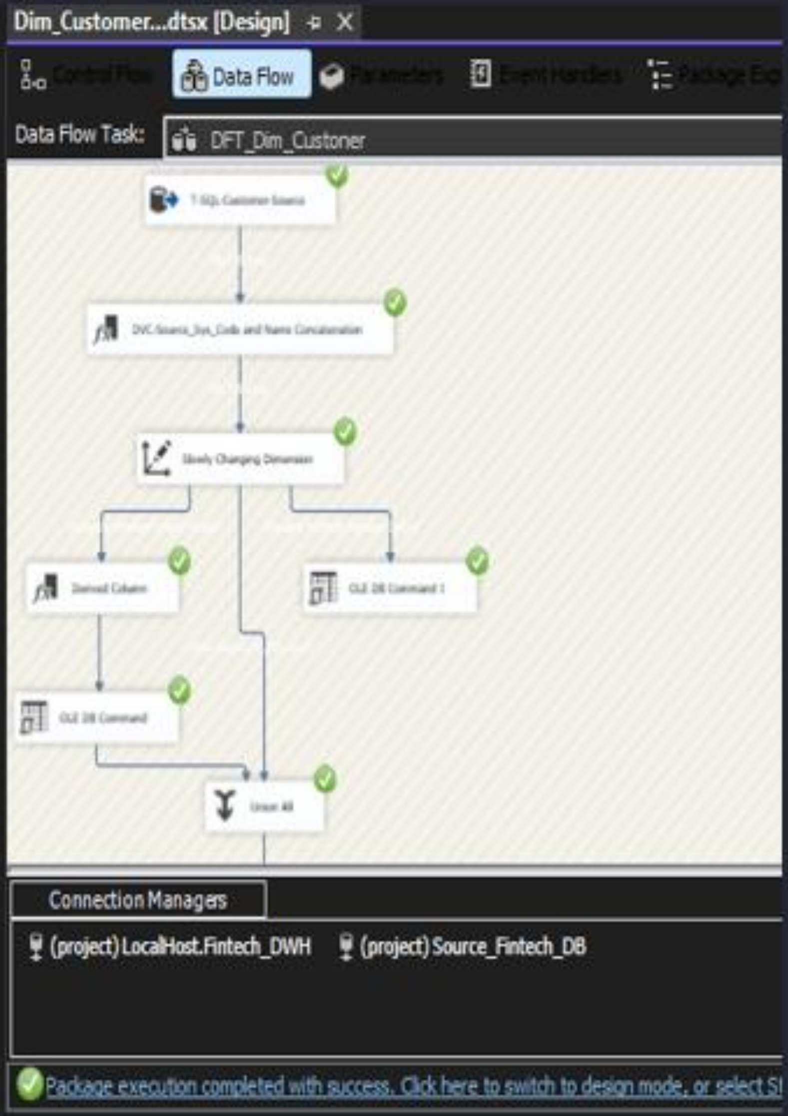
Data Warehouse Design

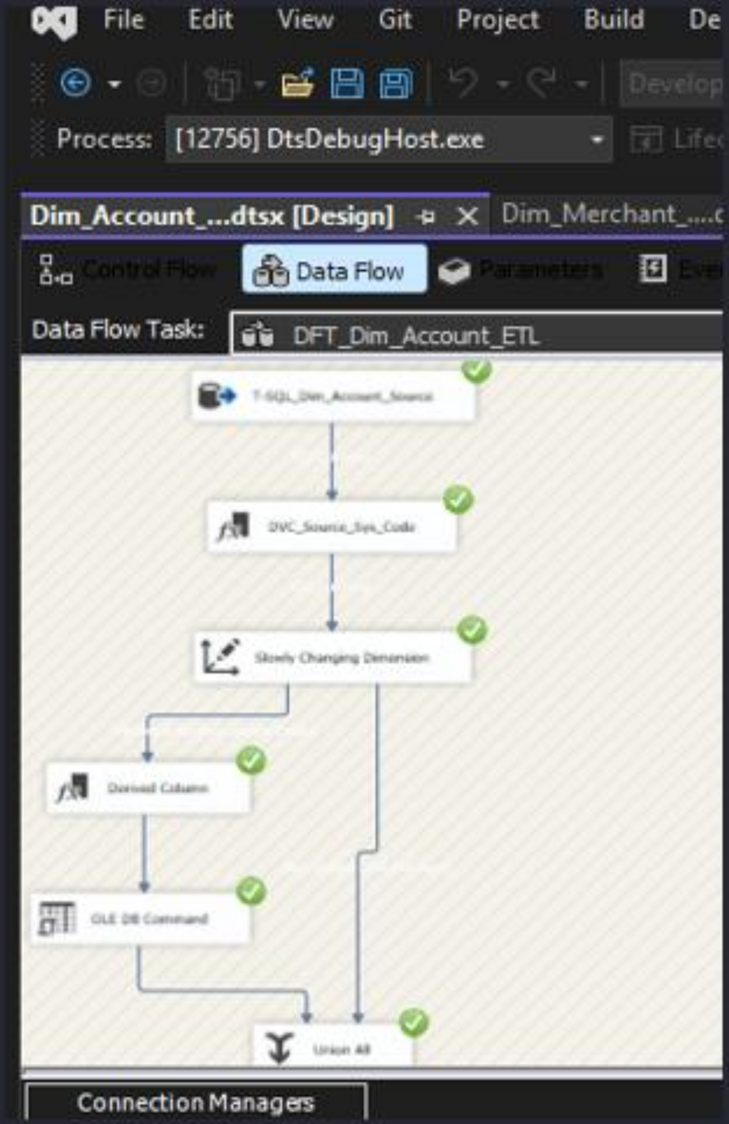


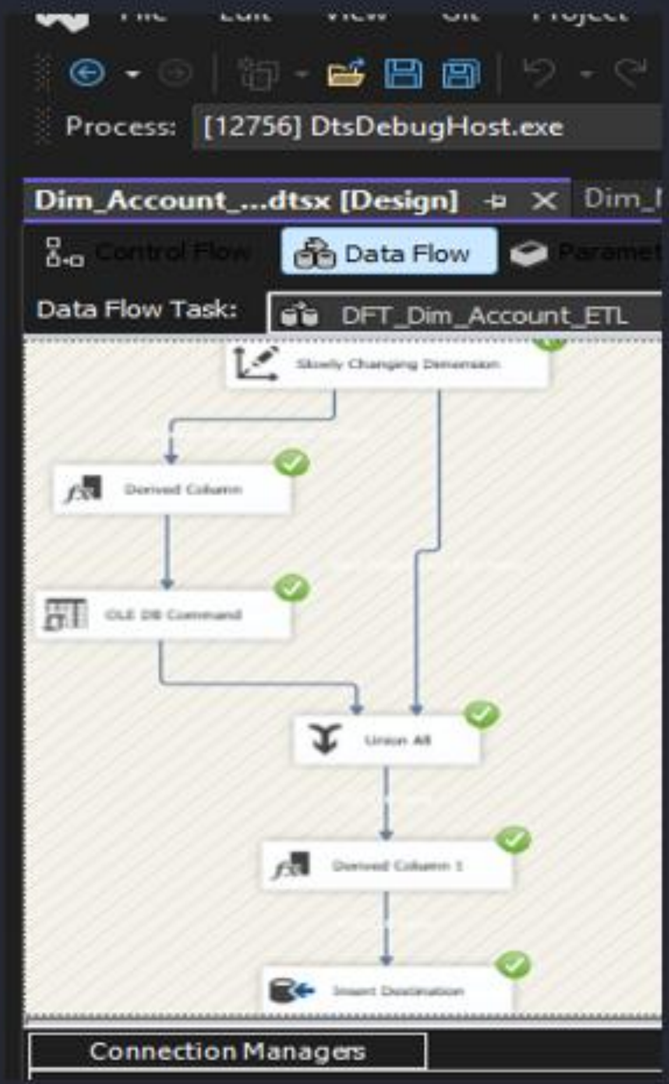
Data Warehouse Schema



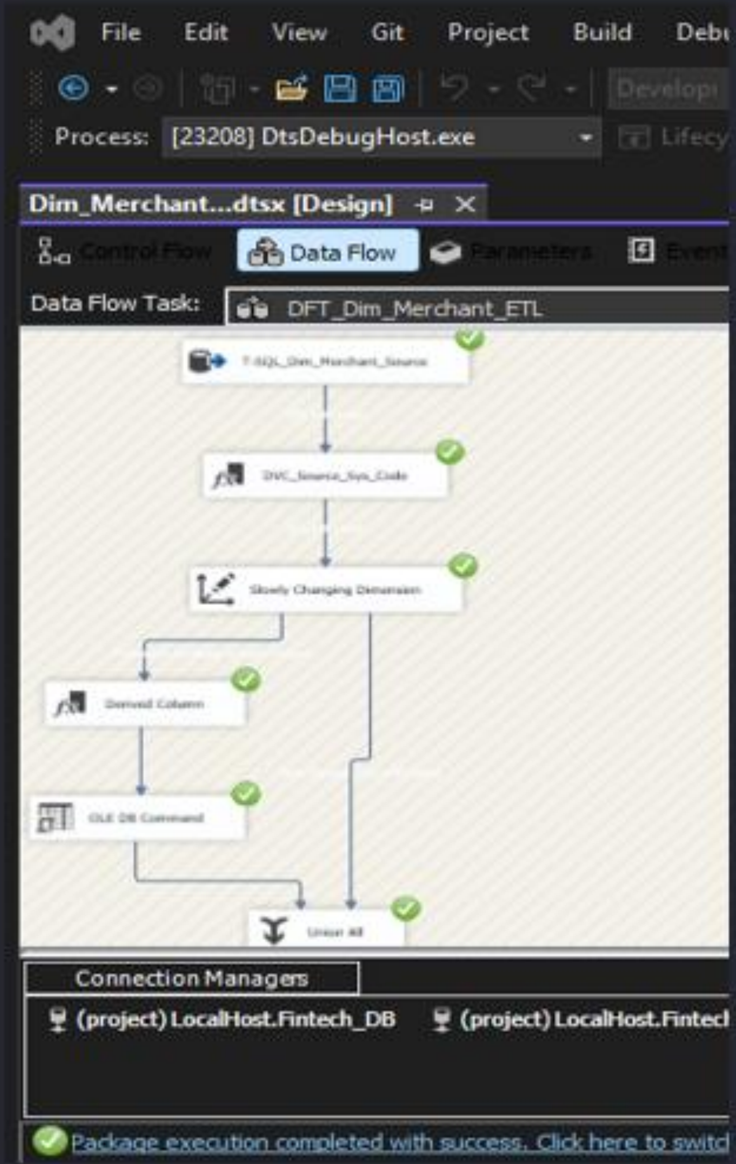
ETL / SSIS: Orchestrating Data Flow

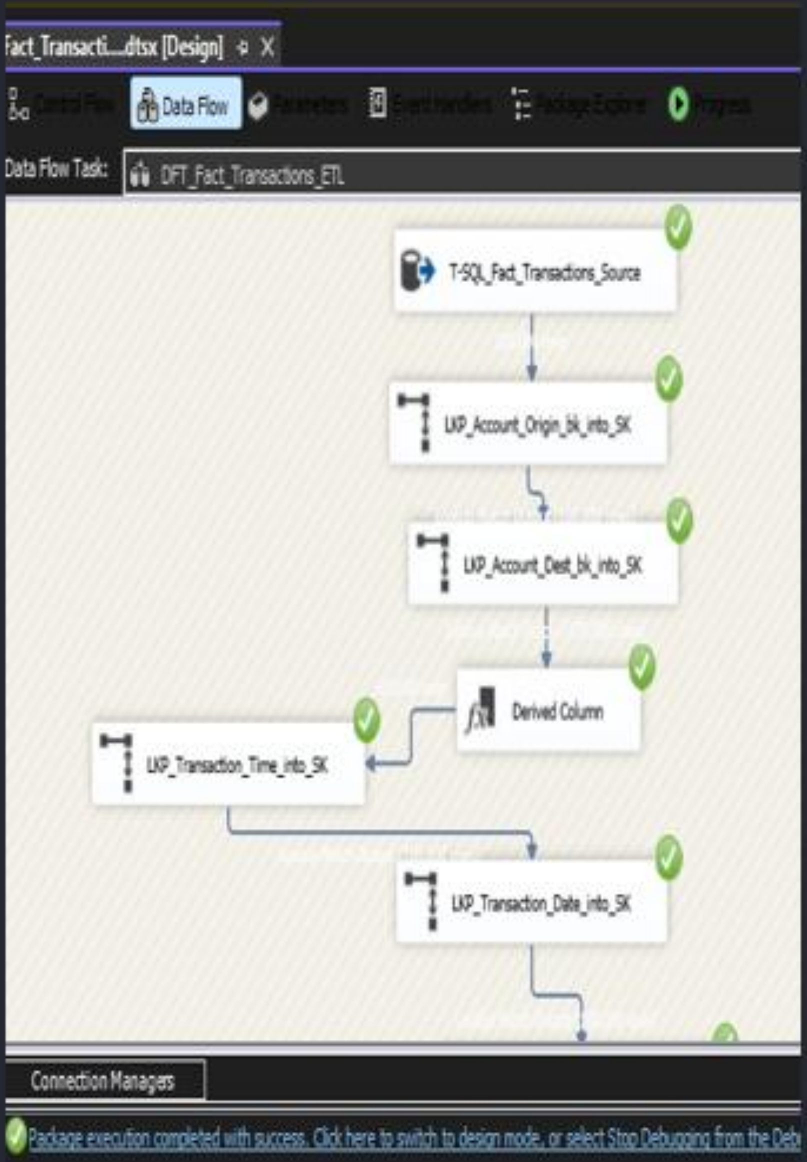


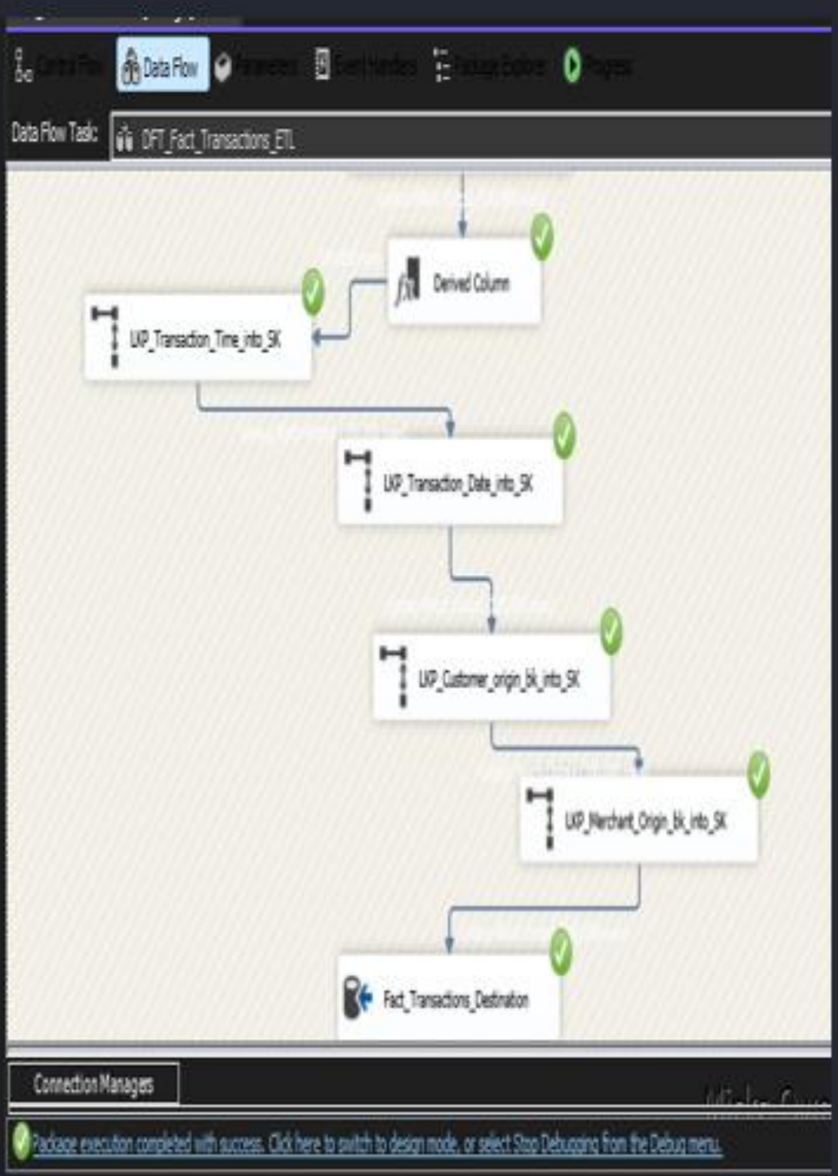




ETL / SSIS: Orchestrating Data Flow







Exploratory Data Analysis: Uncovering Fraud Patterns

1

SQL-Powered Insights

Leveraged advanced SQL queries to extract critical insights from raw data, forming the foundation of our analysis.

2

Advanced Metric Calculation

Utilized complex SQL constructs like Joins, Group By, and Aggregates for precise metric computation. Employed CTEs, Subqueries, and Window Functions for sophisticated ranking and top-N analyses.

3

Comprehensive Fraud Measures

Defined and calculated key fraud metrics, including detected fraud, missed fraud, and flagged fraud incidents, to provide a holistic view.

4

Model Evaluation & Drivers

These detailed insights are crucial for evaluating model performance, identifying top fraud drivers, and informing strategic adjustments.

```sql
--Generate a report that shows all key metrics of the business

select 'total_amount' as measure_name, sum(amount) as measure_value
```

100 %

Results | Messages

	measure_name	measure_value
1	total_amount	17360222176.41
2	number_of_detected_fraud_transactions	2959
3	number_of_fraud_transactions	3063
4	number_of_fraud_transactions_detected_fraud	1959
5	number_of_fraud_not_detected_fraud	1104

```sql
--For each risk segment, rank customers by total fraud amount, and find the top 2 senders per risk segment.
select * from
(select risk_segment, customer_name, sum(amount) as Total_Fraud_Amount,
ROW_NUMBER() over(partition by risk_segment order by sum(amount) desc) as rn
from Fact_Transactions t join Dim_Customer c
on t.Customer_Sender_id_FK = c.Customer_SK
where Is_Fraud = 1
group by Risk_Segment, Customer_Name) as new_table
where rn <= 2
order by Risk_Segment

--Find the top 3 channels with the highest total fraud amount, then calculate their percentage of total fraud
with t as (
```

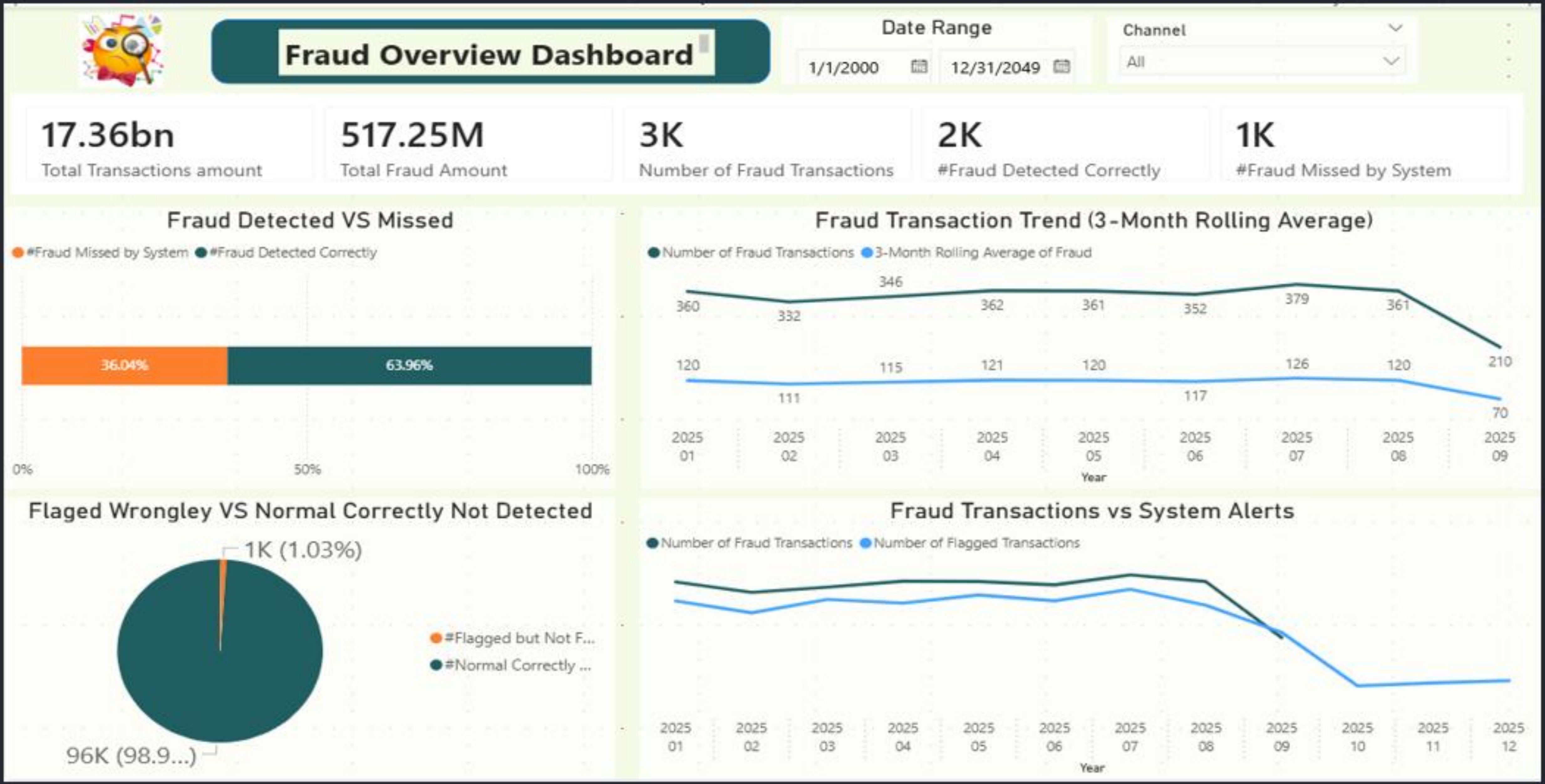
100 %

Results | Messages

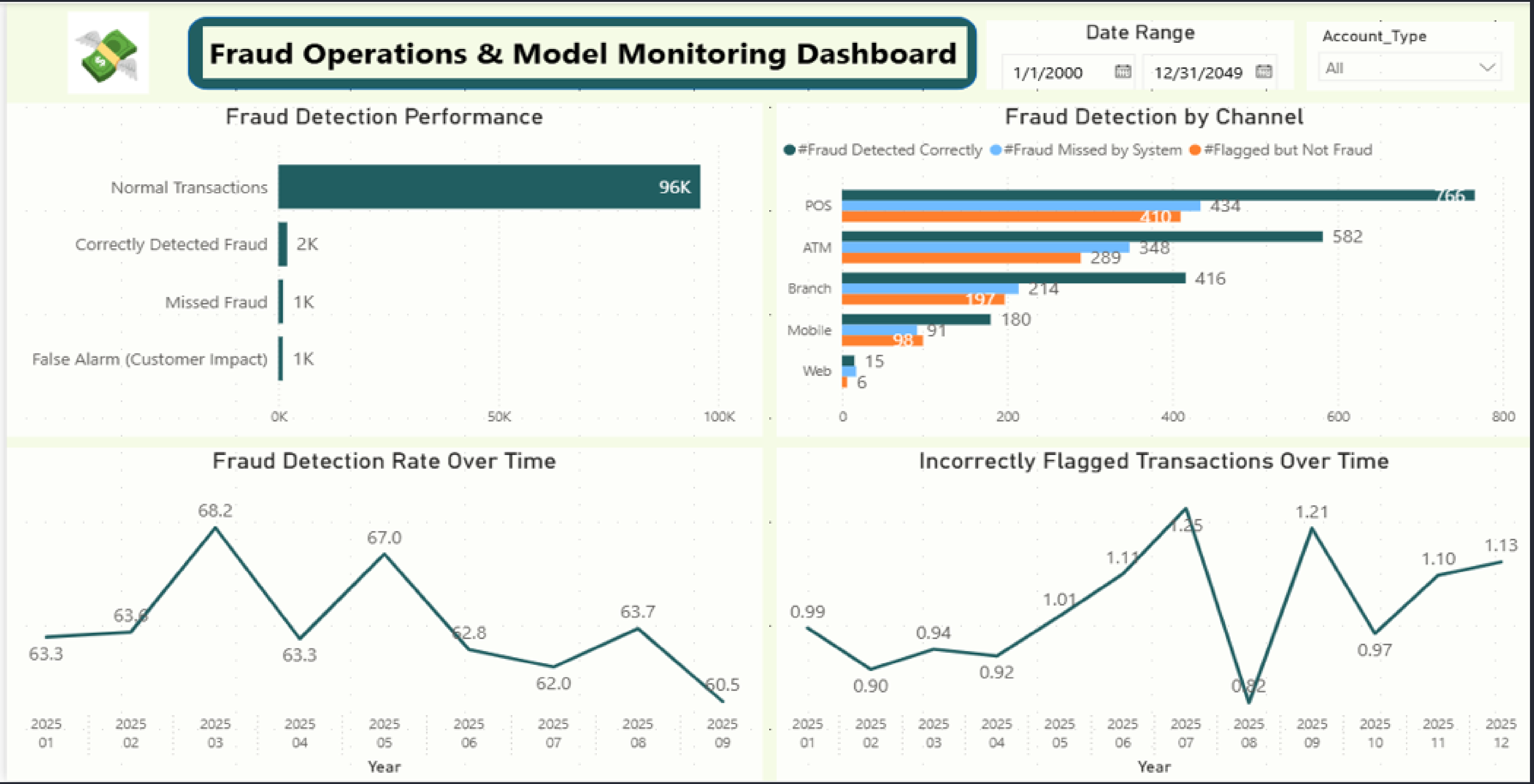
	risk_segment	customer_name	Total_Fraud_Amount	rn
1	High Risk	Linda Kennedy	2745586.7	1
2	High Risk	Jason Tucker	525884.72	2
3	Low Risk	Sherri Morgan	2224025.07	1
4	Low Risk	Andrea Townsend	1902314.85	2
5	Medium Risk	Steven Brown	1053311.47	1
6	Medium Risk	Robert Rodriguez	971258.64	2
7	VIP	Randall Edwards	1547567.06	1
8	VIP	Kristen Yates	1146473.97	2

Power BI Dashboards

FRAUD OVERVIEW Dashboard: Executive Summary



Fraud Operations & Model Monitoring Dashboard

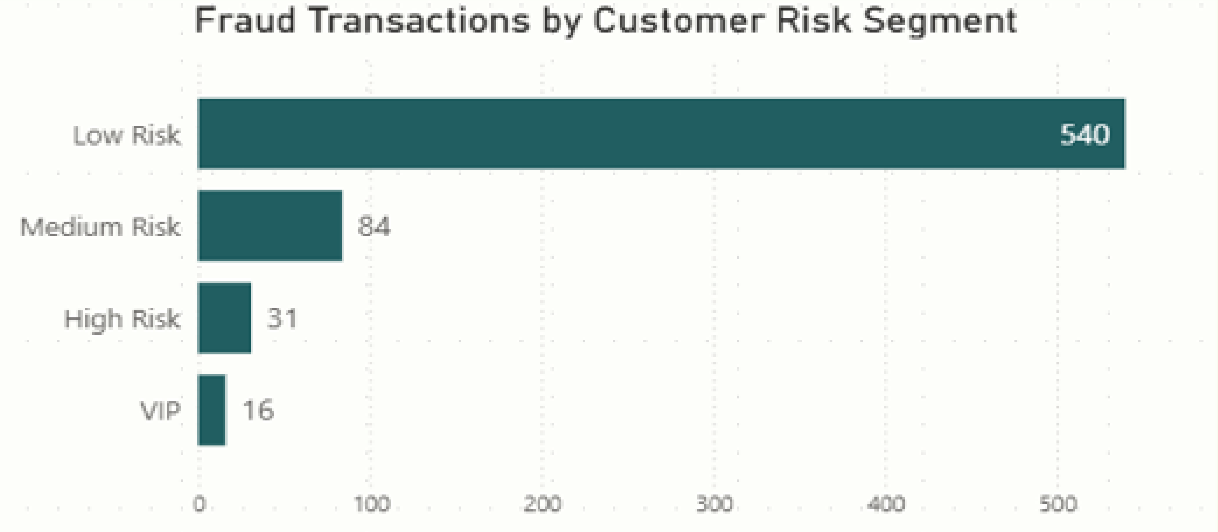


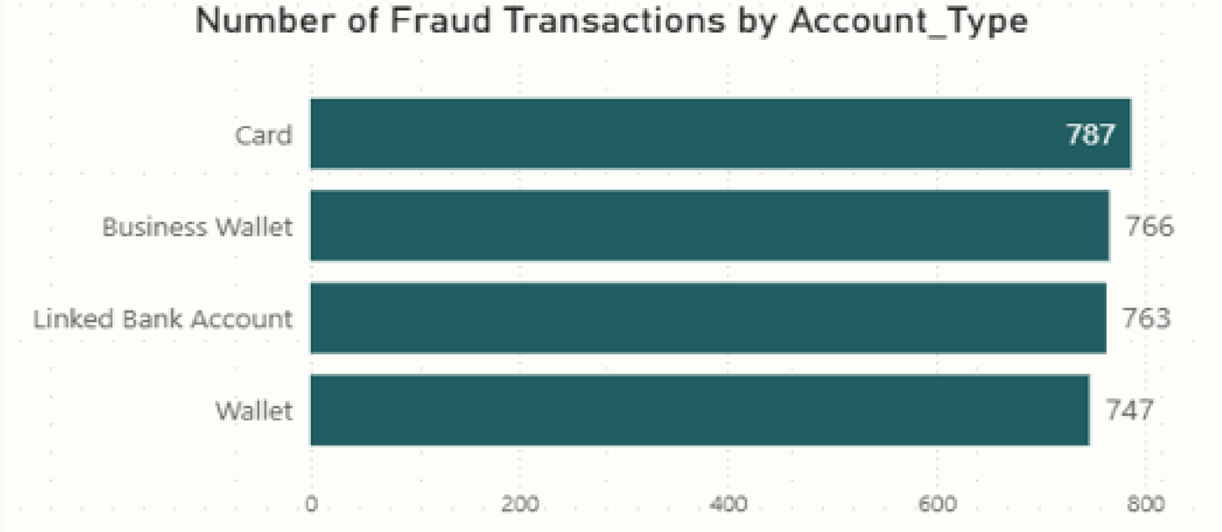
Fraud Drivers & Risk Segmentation Dashboard

Fraud Drivers & Risk Segmentation Dashboard

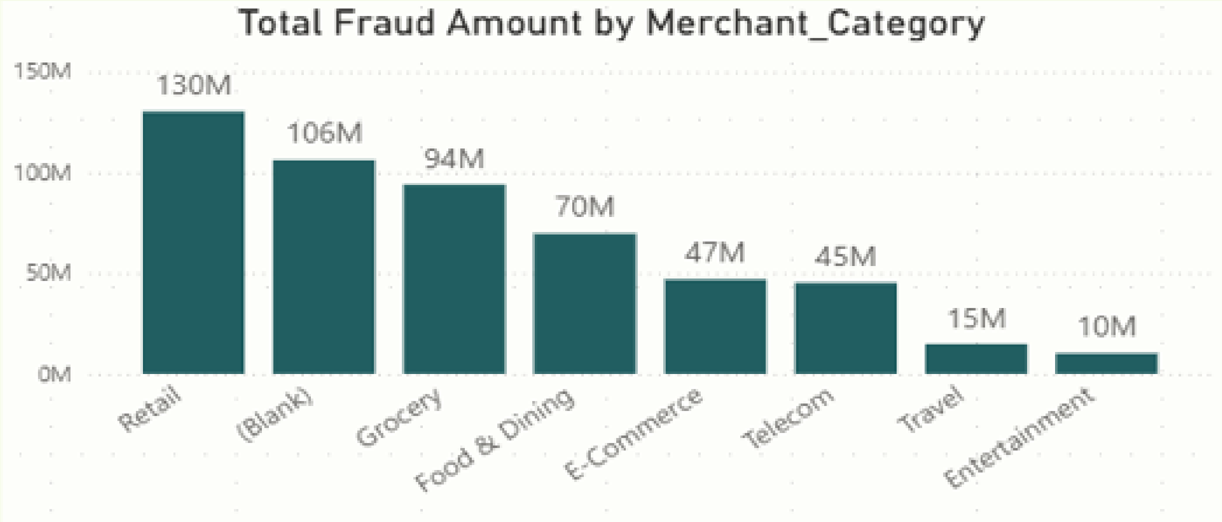
Date Range
1/1/2000 – 12/31/2049

Merchant_Category
All





Merchant_Name	Total Fraud Amount
	106,087,199.87
Little LLC	3,989,315.70
Graves, Holloway and Robinson	3,330,241.40
Allen, Smith and Valdez	3,098,075.96
Joyce and Sons	2,977,893.41
Clay-Hart	2,967,759.64
Nicholson Group	2,803,531.87
Atkinson-Taylor	2,772,552.82
Total	**133,458,684.60**



Key Insights from Fraud Analysis

1. **Enhanced Detection Rate**

   My analysis reveals the precise effectiveness of the fraud detection system, quantifying its ability to identify and prevent fraudulent activities.

2. **Identified Fraud Channels**

   I have successfully identified the top channels, merchants, and customer segments most susceptible to fraud, allowing for targeted interventions.

3. **Actionable Business Intelligence**

   These comprehensive insights are now primed for direct integration into business decision-making, enabling proactive risk management and strategic resource allocation.

Conclusion: A Robust BI Solution for Fraud Detection

This project stands as a testament to a complete, end-to-end Business Intelligence workflow specifically tailored for fraud detection.

01

Integrated Data Flow

Seamlessly integrates data generation, robust database management, efficient ETL processes, and comprehensive EDA.

02

Impactful Dashboards

Culminates in dynamic Power BI dashboards that provide actionable insights to all stakeholders.

03

Ready for Scale

The entire solution is designed for scalability, capable of handling real-world data volumes and evolving fraud landscapes.